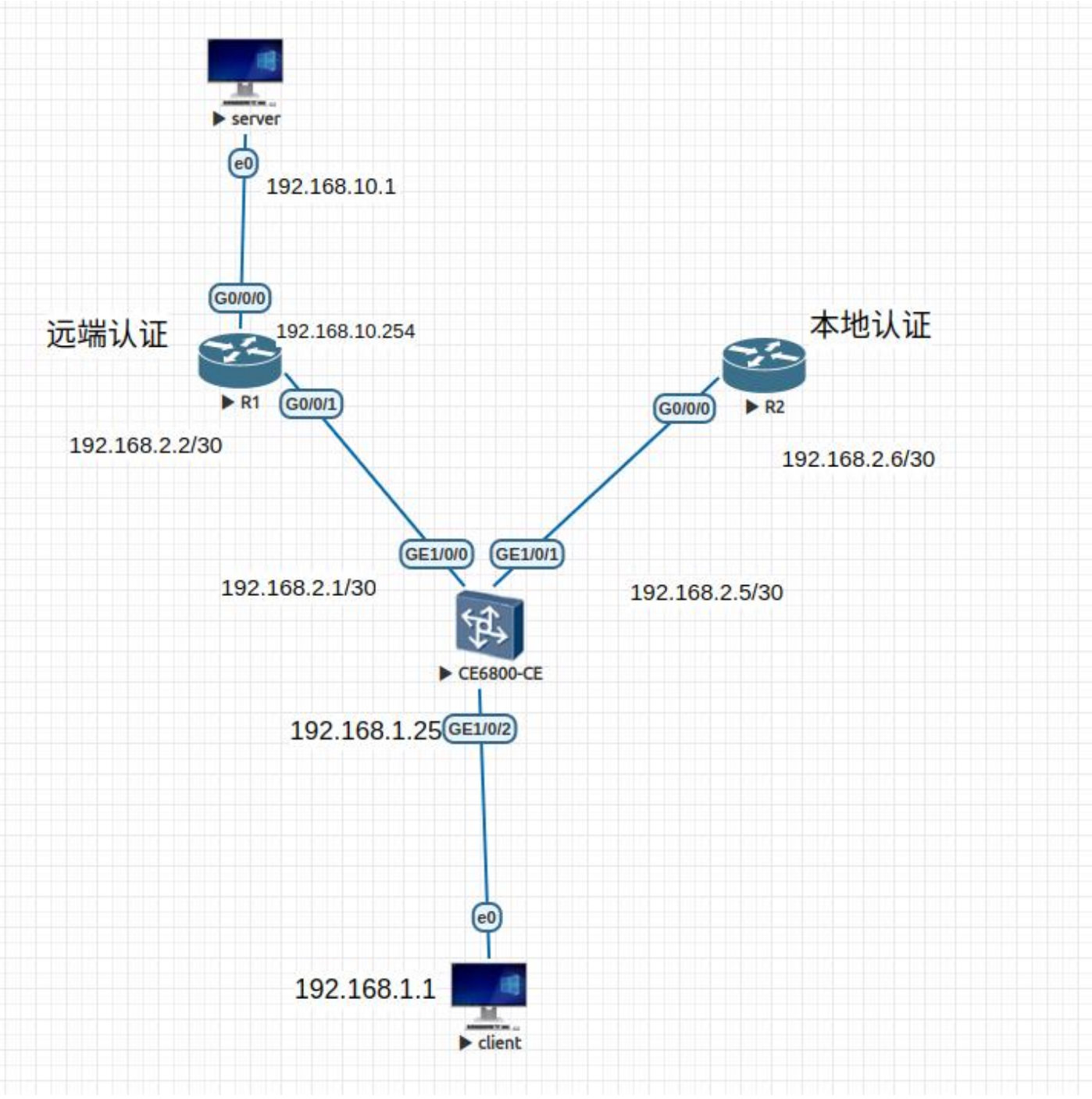
任务 2：AAA 的规划与配置

在企业网络环境中，为实现对用户接入网络的精准管理、安全认证及合理计费，需部署 AAA（认证、授权、计费）服务。本任务基于真实企业网络架构，要求通过配置 RADIUS 服务器与华为路由器，构建一套完整的 AAA 认证与计费体系。



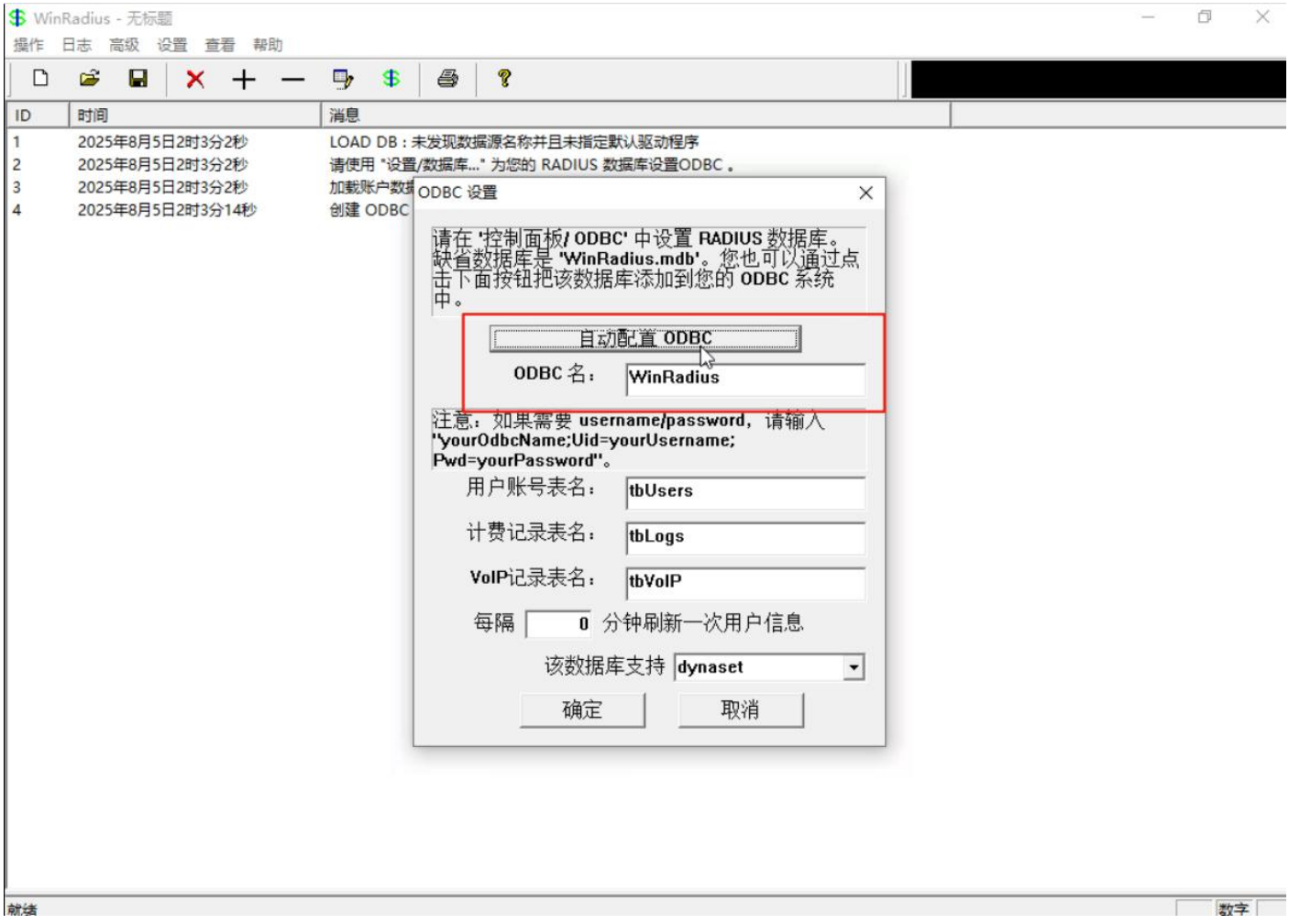
（用户名：admin　密码：Key-1122）

【任务实施】

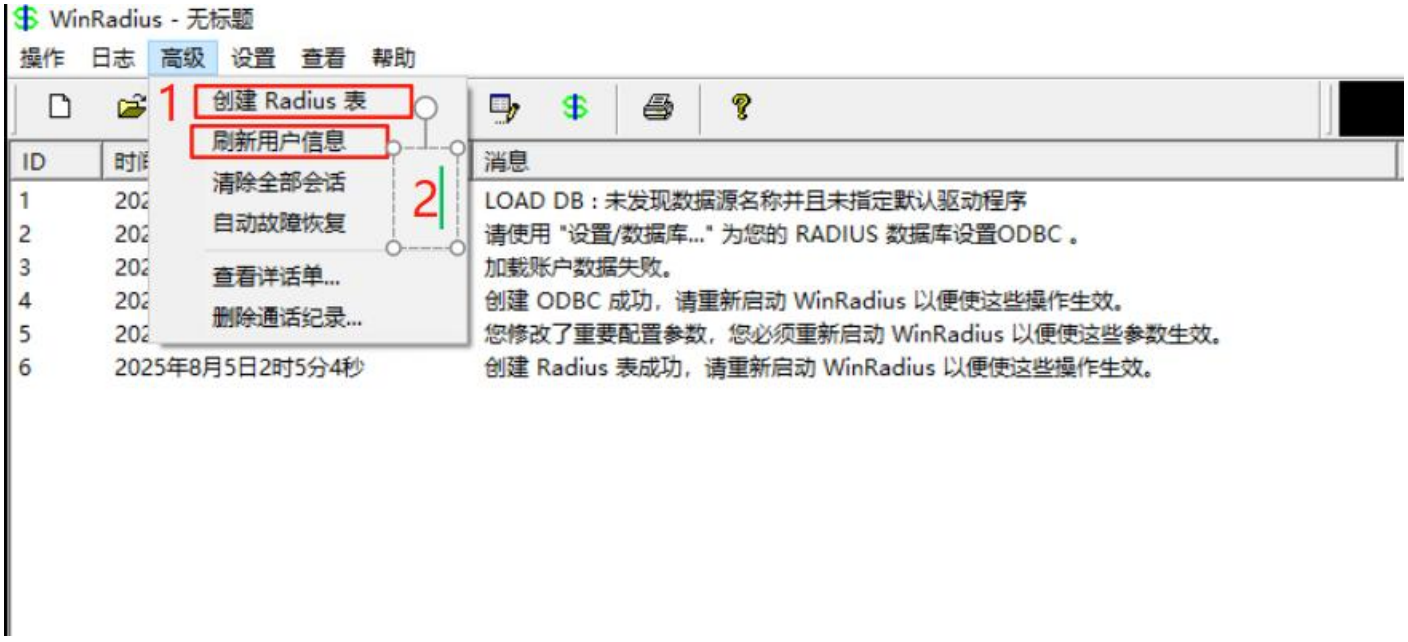
一. 配置 radius 服务器参数

1. 管理员身份打开 radius 服务端

2. 管理员身份运行 WinRadius.exe，设置-->数据库-->自动配置 ODBC 数据库

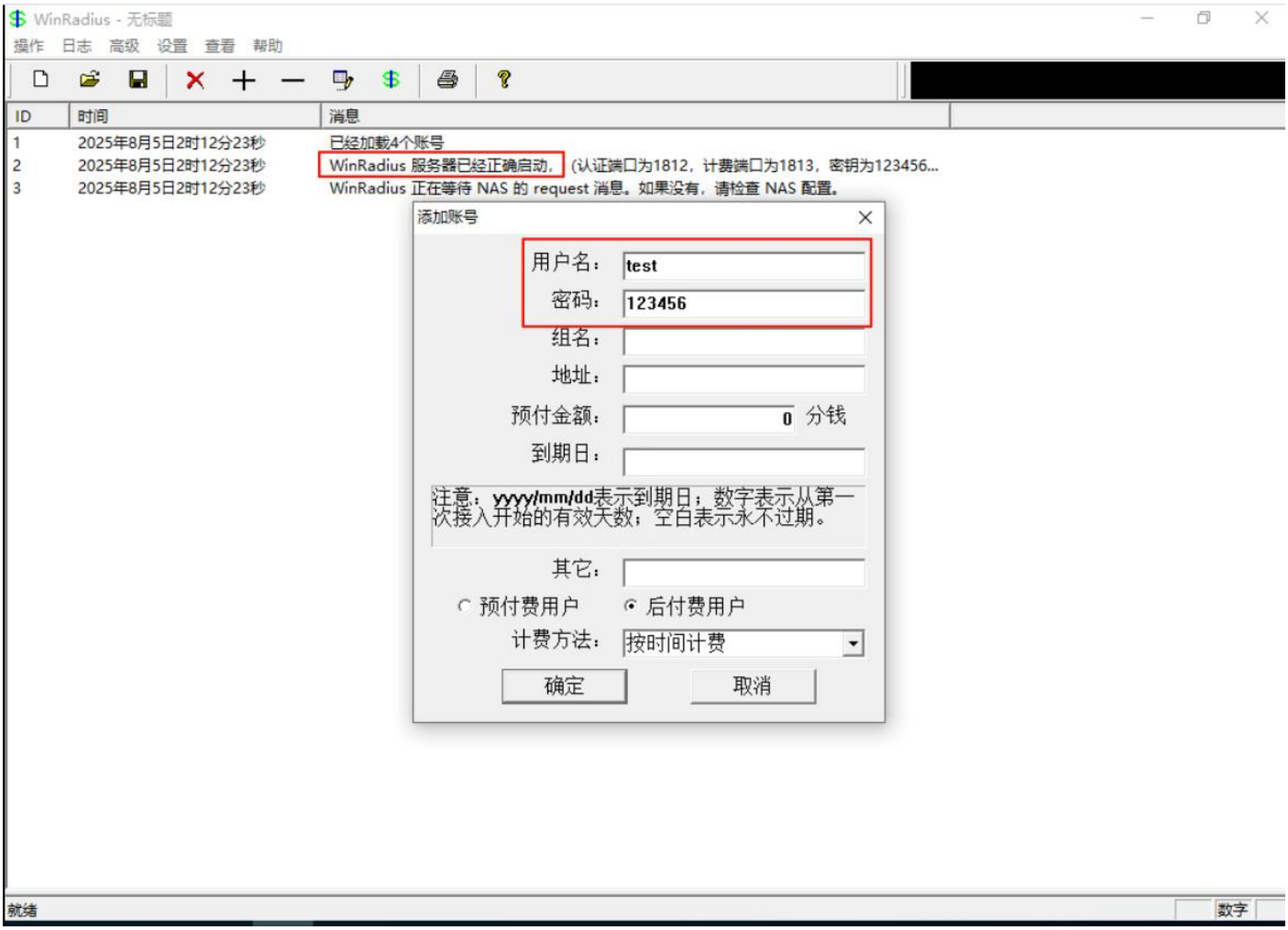


3. 选择高级-->创建 Radius 表

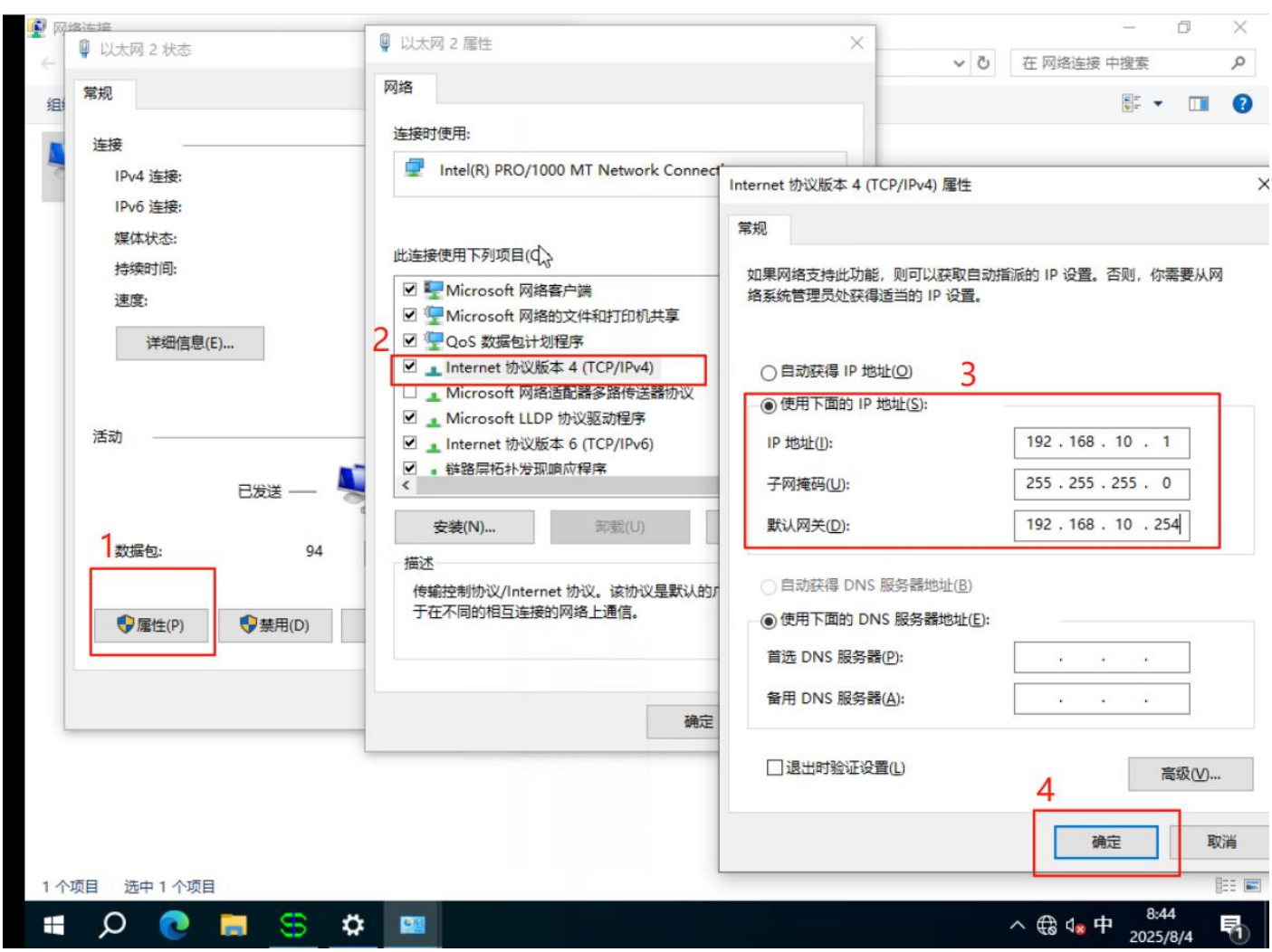


4. 设置共享密钥，选择设置-->系统 输入你需要配置的 NAS 密钥 123123（需与交换机一致）

5. 退出重新打开软件，点击操作，添加账号，输入用户名密码即可添加成功，若是显示添加失败可以先删除再重新添加



6. 配置服务端 IP 地址



二. 基础配置

1. 配置路由器 RT1

（1）登录路由器

```
Username:
Username:super   账号: super
Password:        密码: super
Warning: The password is already expired.
The password needs to be changed. Change now? [Y/N]: Y   选择修改密码
Please enter old password:   输入旧的密码
Please enter new password:   重复输入新的密码
Please confirm new password:
The password has been changed successfully.
<Huawei>
Aug  4 2025 08:49:41+00:00 Huawei AAA/4/ChangePasswordAlarm:OID 1.3.6.1.4.1.2011
<Huawei>
Aug  4 2025 08:49:41+00:00 Huawei LINE/4/USERLOGIN:OID 1.3.6.1.4.1.2011.5.25.207
<Huawei>
<Huawei>sy
Enter system view, return user view with Ctrl+Z.
[Huawei]
```

（2）配置 IP 地址

```
[Huawei]
[Huawei]sysname RT1
[RT1]
[RT1]int g0/0/0
[RT1-GigabitEthernet0/0/0]ip address 192.168.10.254 24
[RT1-GigabitEthernet0/0/0]
Sep  5 2025 02:24:21+00:00 RT1 %%01RM/4/ROUTERID_CHANGE(l)[0]:The
 router ID is 192.168.10.254. (InstanceID=0)
[RT1-GigabitEthernet0/0/0]
Sep  5 2025 02:24:21+00:00 RT1 %%01IFNET/4/LINK_STATE(l)[1]:The l
ine protocol IP on the interface GigabitEthernet0/0/0 has entered
 the UP state.
[RT1-GigabitEthernet0/0/0]
[RT1-GigabitEthernet0/0/0]quit
[RT1]
[RT1]int g0/0/1
[RT1-GigabitEthernet0/0/1]
[RT1-GigabitEthernet0/0/1]ip add 192.168.2.2 30
[RT1-GigabitEthernet0/0/1]
```

（3）配置静态路由

```
[RT1]
[RT1]ip route-static 192.168.1.0 24 192.168.2.1
[RT1]
```

2. 配置路由器 R2

```
[RT2]
[RT2]INT G0/0/0
[RT2-GigabitEthernet0/0/0]
[RT2-GigabitEthernet0/0/0]ip add 192.168.2.6 30
[RT2-GigabitEthernet0/0/0]
Sep  5 2025 02:30:26+00:00 RT2 %%01RM/4/ROUTERID_CHANGE(l)[0]:The
 router ID is 192.168.2.6. (InstanceID=0)
[RT2-GigabitEthernet0/0/0]
Sep  5 2025 02:30:26+00:00 RT2 %%01IFNET/4/LINK_STATE(l)[1]:The l
ine protocol IP on the interface GigabitEthernet0/0/0 has entered
 the UP state.
[RT2-GigabitEthernet0/0/0]
[RT2-GigabitEthernet0/0/0]quit
[RT2]
[RT2]ip route-static 192.168.1.0 24 192.168.2.5
[RT2]
```

3. 配置交换机

```
[HUAWEI]vlan batch 10 100 200
[HUAWEI]
[HUAWEI]int vlanif 10
[HUAWEI-Vlanif10]
[HUAWEI-Vlanif10]ip add 192.168.1.254 24
[HUAWEI-Vlanif10]
[HUAWEI-Vlanif10]int vlanif 100
[HUAWEI-Vlanif100]
[HUAWEI-Vlanif100]ip add 192.168.2.1 30
[HUAWEI-Vlanif100]
[HUAWEI-Vlanif100]int vlanif 200
[HUAWEI-Vlanif200]
[HUAWEI-Vlanif200]ip add 192.168.2.5 30
[HUAWEI-Vlanif200]
```

```
[HUAWEI]
[HUAWEI]interface GE 1/0/2
[HUAWEI-GE1/0/2]
[HUAWEI-GE1/0/2]port default vlan 10
[HUAWEI-GE1/0/2]
[HUAWEI-GE1/0/2]interface GE 1/0/0
[HUAWEI-GE1/0/0]
[HUAWEI-GE1/0/0]port default vlan 100
[HUAWEI-GE1/0/0]
[HUAWEI-GE1/0/0]interface GE 1/0/1
[HUAWEI-GE1/0/1]
[HUAWEI-GE1/0/1]port default vlan 200
[HUAWEI-GE1/0/1]
[HUAWEI-GE1/0/1]quit
[HUAWEI]
[HUAWEI]ip route-static 192.168.10.0 24 192.168.2.2
[HUAWEI]
```

三. RT1 配置 AAA 认证(远端认证)

1. 配置 RADIUS 认证服务器和计费服务器的 IP 地址、端口。

```
[RT1]
[RT1]radius-server template gzqk
[RT1-radius-gzqk]
[RT1-radius-gzqk]radius-server authentication 192.168.10.1 1812   # 指定认证服务器ip地址
[RT1-radius-gzqk]
[RT1-radius-gzqk]radius-server accounting 192.168.10.1 1813   # 指定计费服务器ip地址
[RT1-radius-gzqk]
[RT1-radius-gzqk]quit
[RT1]
```

2. 配置 RADIUS 服务器密钥、重传次数，以及设备向 RADIUS 服务器发送的报文中的用户名不包含域名。

```
[RT1]
[RT1]radius-server template gzqk
[RT1-radius-gzqk]
[RT1-radius-gzqk]radius-server shared-key cipher 123456   # 配置radius服务器共享密钥
Warning: The shared-key complexity is low. A more secure shared-key is recommended. Continue?[Y/N]y
[RT1-radius-gzqk]
[RT1-radius-gzqk]radius-server retransmit 2   # 设置重传次数
[RT1-radius-gzqk]
[RT1-radius-gzqk]
[RT1-radius-gzqk]undo radius-server user-name domain-included   # 设置用户名不包括域名
[RT1-radius-gzqk]
[RT1-radius-gzqk]quit
[RT1]
```

3. 配置认证方案 auth

```
[RT1]
[RT1]aaa
[RT1-aaa]
[RT1-aaa]authentication-scheme auth   # 配置AAA认证方案
[RT1-aaa-authen-auth]
[RT1-aaa-authen-auth]authentication-mode radius
[RT1-aaa-authen-auth]   # 设置认证模式为radius
[RT1-aaa-authen-auth]quit
[RT1-aaa]
```

4. 配置计费方案 abc，计费模式为 RADIUS，并配置当开始计费失败时，允许用户上线。

```
[RT1-aaa]
[RT1-aaa]accounting-scheme abc                    # 配置计费方案
[RT1-aaa-accounting-abc]
[RT1-aaa-accounting-abc]accounting-mode radius     # 配置计费模式为radius
[RT1-aaa-accounting-abc]
[RT1-aaa-accounting-abc]accounting start-fail offline
[RT1-aaa-accounting-abc]
[RT1-aaa-accounting-abc]
```

5. 配置 huawei 域，在域下应用认证方案 auth、计费方案 abc、RADIUS 服务器模板 gzqk

```
[RT1-aaa]
[RT1-aaa]
[RT1-aaa]domain huawei    # 设置域名称
[RT1-aaa-domain-huawei]
[RT1-aaa-domain-huawei]authentication-scheme auth    # 应用认证方案
[RT1-aaa-domain-huawei]
[RT1-aaa-domain-huawei]accounting-scheme abc         # 应用计费方案
[RT1-aaa-domain-huawei]
[RT1-aaa-domain-huawei]radius-server gzqk            # 应用radius服务器模板
[RT1-aaa-domain-huawei]
```

四. R2 配置本地认证

1. 配置 AAA 本地认证。

```
[RT2]aaa
[RT2-aaa]
[RT2-aaa]local-user admin password cipher gzqk@123     # 创建本地用户名和密码
Info: Add a new user.
[RT2-aaa]
[RT2-aaa]local-user admin privilege level 15          # 设置admin用户权限等级
Info: After changing the rights (including the password, access type, FTP directory, bind IP, and level)
 users already online do not change. The change takes effect to users who go online after the change.
[RT2-aaa]
[RT2-aaa]local-user admin service-type telnet  # 配置本地用户可以通过 Telnet 方式来访问和管理设备
Info: The cipher password has been changed to an irreversible-cipher password.
Warning: The user access modes include Telnet, FTP, or HTTP, so security risks exist.
Info: After changing the rights (including the password, access type, FTP directory, bind IP, and level)
 users already online do not change. The change takes effect to users who go online after the change.
[RT2-aaa]
```

五. 验证配置结果

1. 在 R1 上执行命令 display radius-server configuration template template-name，可以观察到该 RADIUS 服务器模板的配置与要求一致。

```
[Huawei]display radius-server configuration template gzqk
  ------------------------------------------------------------------------------
  Server-template-name          :  gzqk
  Protocol-version              :  standard
  Traffic-unit                  :  B
  Shared-secret-key             :  %^%#`3:d(.fJI0eQUj:.'U=J.5}%.#CS[IoAy#2y1jN5%^%#
  Group-filter                  :  class
  Timeout-interval(in second)   :  5
  Retransmission                :  2
  EndPacketSendTime             :  3
  Dead time(in minute)          :  5
  Domain-included               :  NO
  NAS-IP-Address                :  -
  Calling-station-id MAC-format :  xxxx-xxxx-xxxx
  Called-station-id MAC-format  :  XX-XX-XX-XX-XX-XX
  NAS-Port-ID format            :  New
  Service-type                  :  -
  NAS-IPv6-Address              :  ::
  Server algorithm              :  master-backup
  Detect-interval(in second)    :  60
  Detect up-server(in second)   :  0
  Detect timeout(in second)     :  3
  Chargeable-user-identity      :  Not Support
  CUI Not reject                :  No
  Authentication Server 1       :  192.168.10.1    Port:1812  Weight:80  [up]
                                   Vrf:- LoopBack:NULL Vlanif:NULL
                                   Source IP: ::
  Accounting Server     1       :  192.168.10.1    Port:1813  Weight:80  [up]
                                   Vrf:- LoopBack:NULL Vlanif:NULL
                                   Source IP: ::
  ------------------------------------------------------------------------------
[Huawei]
```

2. R1 查看域信息

```
[Huawei]display domain name huawei

  Domain-name                      : huawei
  Domain-index                     : 2
  Domain-state                     : Active
  Authentication-scheme-name       : auth
  Accounting-scheme-name           : abc
  Authorization-scheme-name        : -
  Service-scheme-name              : -
  RADIUS-server-template           : gzqk
  Accounting-copy-RADIUS-template  : -
  HWTACACS-server-template         : -
  User-group                       : -
  Push-url-address                 : -
  Flow-statistic                   : -
  Tariff-level                     : -
```

3. 客户端主机配置地址网关与路由器 0/0/1 接口地址互联。

4. 使用 radius 客户端软件认证 radius 服务器。